SCENIC

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Loading packages

```
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(patchwork))
suppressMessages(library(ggplot2))
suppressMessages(library(ComplexHeatmap))
suppressMessages(library(SCENIC))
suppressMessages(library(stringr))
suppressMessages(library(tidyHeatmap))
```

Loading data

```
Cell_HiveAnnotate <- readRDS("../Hive-2-Annotation/lavage_Hive_annotated.rds")
```

prepare cellinfo et expression matrix

```
exprMat <- Cell_HiveAnnotate@assays$RNA@data
cellInfo <- data.frame(seuratCluster=Idents(Cell_HiveAnnotate))
cellInfo$Annotation <- Cell_HiveAnnotate$Annotation
cellInfo$nGene <- colSums(exprMat>0)
```

```r
dir.create ("int")
saveRDS(exprMat, file = "int/exprMat.Rds")
saveRDS(cellInfo, file = "int/cellInfo.Rds")
```

```r
cellInfo <- readRDS(file = "int/cellInfo.Rds")
exprMat <- readRDS(file = "int/exprMat.Rds")
```

Download binding motifs for HUMAN

```r
dbFiles <-
c("https://resources.aertslab.org/cistarget/databases/old/homo_sapiens/hg19/refseq_r45/mc9nr/gene_based/
"https://resources.aertslab.org/cistarget/databases/old/homo_sapiens/hg19/refseq_r45/mc9nr/gene_based/hg

dir.create("cisTarget_databases");
setwd("cisTarget_databases")
for(featherURL in dbFiles)
{
  download.file(featherURL, destfile=basename(featherURL)) # saved in current dir
}
```

Initialise Scenic

```r
data(list="motifAnnotations_hgnc_v9", package="RcisTarget")
motifAnnotations_hgnc <- motifAnnotations_hgnc_v9

org <- "hgnc"
dbDir <- "cisTarget_databases"
dbDir <- path.expand(dbDir)
myDatasetTitle <- "SCENIC Analysis"
data(defaultDbNames)
dbs <- defaultDbNames[[org]]
scenicOptions <- initializeScenic(org=org, dbDir=dbDir, dbs=dbs, nCores=25)
```

```
## Motif databases selected:
##   hg19-500bp-upstream-7species.mc9nr.feather
##   hg19-tss-centered-10kb-7species.mc9nr.feather
```

```
## Using the column 'features' as feature index for the ranking database.
## Using the column 'features' as feature index for the ranking database.
```

2. Filter by the number of cells in which the gene is detected.

```r
exprMat <- as.matrix(exprMat) # need a regulat matrix
genesKept <- geneFiltering(exprMat, scenicOptions=scenicOptions,
                           minCountsPerGene=3*.01*ncol(exprMat),
                           minSamples=ncol(exprMat)*.01)

dim(exprMat) # 34235
exprMat_filtered <- exprMat[genesKept, ]
dim(exprMat_filtered) # 8322
```

Correlation

```
runCorrelation(exprMat_filtered, scenicOptions)
```

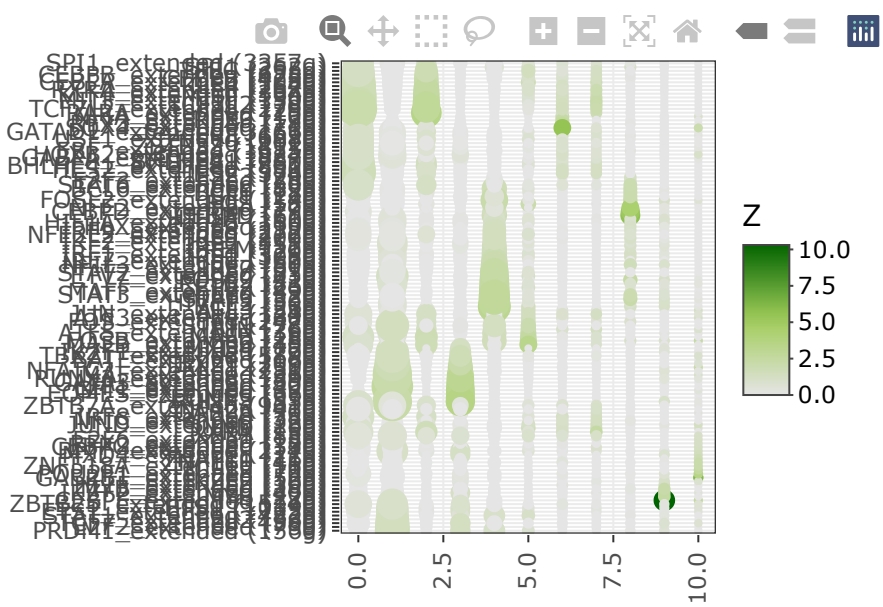
```
runGenie3(exprMat_filtered, scenicOptions)
```

```
scenicOptions@settings$verbose <- TRUE
scenicOptions@settings$nCores <- 10
scenicOptions@settings$seed <- 123

runSCENIC_1_coexNetwork2modules(scenicOptions)
runSCENIC_2_createRegulons(scenicOptions)
runSCENIC_3_scoreCells(scenicOptions, exprMat_filtered)
```

Plotting the results

Viewing markers based on specificity score

```
regulonAUC <- loadInt(scenicOptions, "aucell_regulonAUC") # require file
int/3.4_regulonAUC.Rds


rss <- calcRSS(AUC=getAUC(regulonAUC), cellAnnotation=cellInfo[colnames(regulonAUC),
"seuratCluster"])
#rssPlot <- plotRSS(rss["Mafb (44g)",])
rssPlot <- plotRSS(rss)
plotly::ggplotly(rssPlot$plot)
```



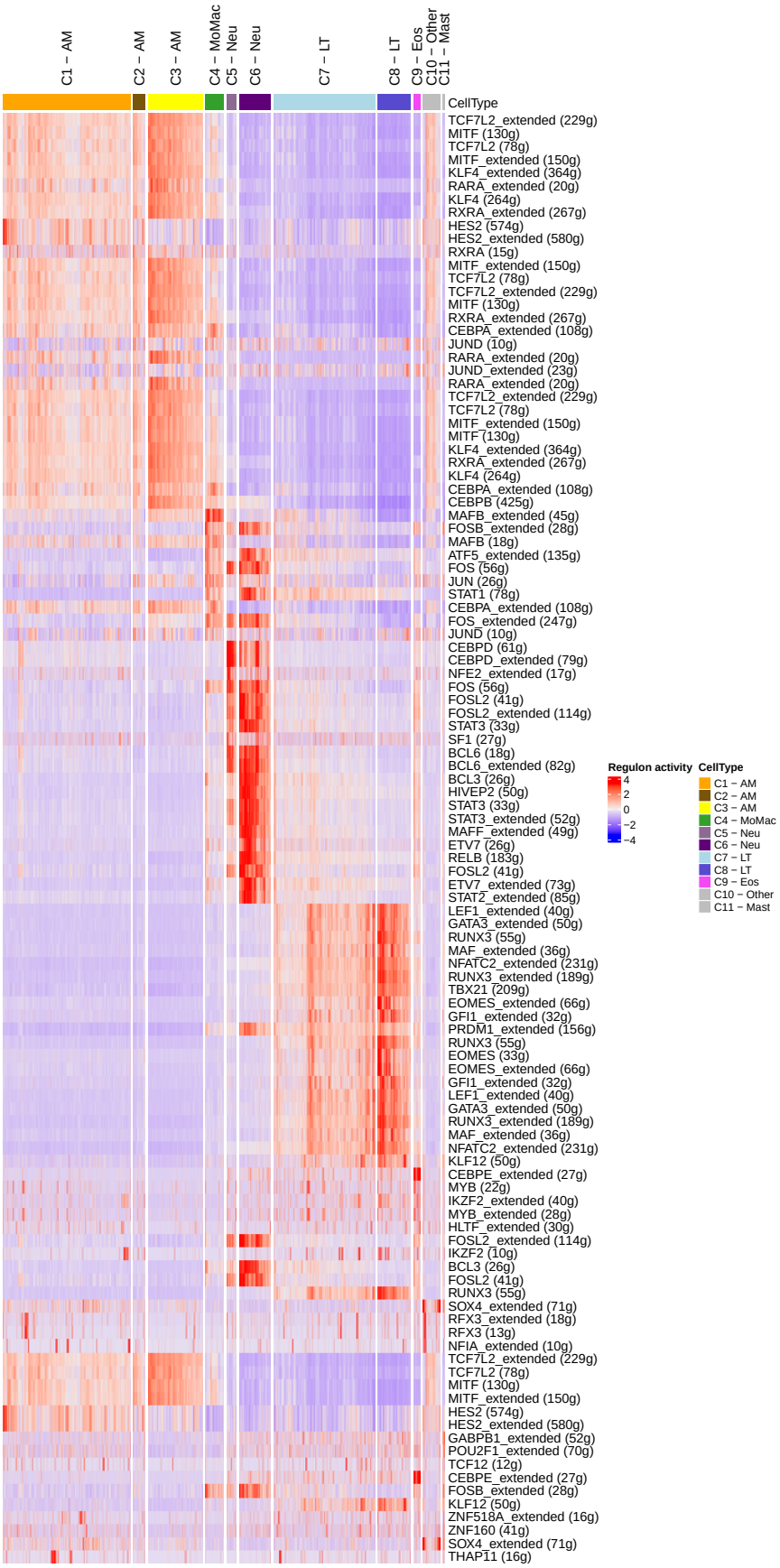
Plot on Heatmap the 10 markers the most specific of each cell type

```
#rss <- rss[,c(8,11,2,3,7,1,12,9,4,6,5,10)]
rss <- rss[,c(3,11,1,7,5,6,2,8,4,10,9)]
```

```r
# J'ai changé l'ordre des clusters donc à modifier
rss <- rss[,c(1,8,3,6,9,5,2,4,10,7,11)]

top_tf <- c()
for (i in 1:length(colnames(rss))) {

  TFs <- sort(rss[,i], decreasing = T)
  top_10 <- head(TFs, 10)
  top_tf <- c(top_tf,top_10)
}

length(top_tf)

top_tf

# Creation of the Heatmap

regulonAUC.mat <- regulonAUC@assays@data@listData$AUC
Subset_regulonActivity <-regulonAUC.mat[names(top_tf),]

regulonActivity_byCellType_Scaled <- t(scale(t(Subset_regulonActivity), center = T,
scale=T))

colors <- c("#ff9305", "#56B4E9", "#009E73", "#F0E442", "#0072B2",
                "#877f5f", "#CC79A7", "#c0bec2", "#854e07", "#a865eb",
                "#00A087", "#FF9DA7")

names(colors) <- levels(Cell_HiveAnnotate$Annotation)

df <- as.data.frame(Cell_HiveAnnotate$Annotation)
colnames(df) <- "CellType"

color_df <- list(CellType =
                   c("C1 - AM" = "orange",
                     "C2 - AM" = "#78550b",
                     "C3 - AM" = "yellow",
                     "C4 - MoMac" = "#33A02C",
                     "C5 - Neu" = "#8c6a94",
                     "C6 - Neu" = "#600078",
                     "C7 - LT" = "lightblue",
                     "C8 - LT" = "#584acf",
                     "C9 - Eos" = "#f249f5",
                     "C10 - Other" = "gray",
                     "C11 - Mast" = "gray"))

Heatmap <- Heatmap(regulonActivity_byCellType_Scaled, name="Regulon activity",
show_column_names = FALSE,
        column_split = factor(Cell_HiveAnnotate$Annotation),
        cluster_column_slices = F,
        cluster_rows = F,
        column_title_rot = 90,
        top_annotation = HeatmapAnnotation(df = df, col = color_df),
```

```r
        show_column_dend = F)

# tidyHeatmap::save_pdf(Heatmap,"Scenic_Hive.pdf", width = 40, height = 50, units = "cm")
```

```r
Heatmap
```


C1 – AM
C2 – AM
C3 – AM
C4 – MoMac
C5 – Neu
C6 – Neu
C7 – LT
C8 – LT
C9 – Eos
C10 – Other
C11 – Mast
CellType
TCF7L2_extended (229g)
MITF (130g)
TCF7L2 (78g)
MITF_extended (150g)
KLF4_extended (364g)
RARA_extended (20g)
KLF4 (264g)
RXRA_extended (267g)
HES2 (574g)
HES2_extended (580g)
RXRA (15g)
MITF_extended (150g)
TCF7L2 (78g)
TCF7L2_extended (229g)
MITF (130g)
RXRA_extended (267g)
CEBPA_extended (108g)
JUND (10g)
RARA_extended (20g)
JUND_extended (23g)
RARA_extended (20g)
TCF7L2_extended (229g)
TCF7L2 (78g)
MITF_extended (150g)
MITF (130g)
KLF4_extended (364g)
RXRA_extended (267g)
KLF4 (264g)
CEBPA_extended (108g)
CEBPB (425g)
MAFB_extended (45g)
FOSB_extended (28g)
MAFB (18g)
ATF5_extended (135g)
FOS (56g)
JUN (26g)
STAT1 (78g)
CEBPA_extended (108g)
FOS_extended (247g)
JUND (10g)
CEBPD (61g)
CEBPD_extended (79g)
NFE2_extended (17g)
FOS (56g)
FOSL2 (41g)
FOSL2_extended (114g)
STAT3 (33g)
SF1 (27g)
BCL6 (18g)
BCL6_extended (82g)
BCL3 (26g)
HIVEP2 (50g)
STAT3 (33g)
STAT3_extended (52g)
MAFF_extended (49g)
ETV7 (26g)
RELB (183g)
FOSL2 (41g)
ETV7_extended (73g)
STAT2_extended (85g)
LEF1_extended (40g)
GATA3_extended (50g)
RUNX3 (55g)
MAF_extended (36g)
NFATC2_extended (231g)
RUNX3_extended (189g)
TBX21 (209g)
EOMES_extended (66g)
GFI1_extended (32g)
PRDM1_extended (156g)
RUNX3 (55g)
EOMES (33g)
EOMES_extended (66g)
GFI1_extended (32g)
LEF1_extended (40g)
GATA3_extended (50g)
RUNX3_extended (189g)
MAF_extended (36g)
NFATC2_extended (231g)
KLF12 (50g)
CEBPE_extended (27g)
MYB (22g)
IKZF2_extended (40g)
MYB_extended (28g)
HLTF_extended (30g)
FOSL2_extended (114g)
IKZF2 (10g)
BCL3 (26g)
FOSL2 (41g)
RUNX3 (55g)
SOX4_extended (71g)
RFX3_extended (18g)
RFX3 (13g)
NFIA_extended (10g)
TCF7L2_extended (229g)
TCF7L2 (78g)
MITF (130g)
MITF_extended (150g)
HES2 (574g)
HES2_extended (580g)
GABPB1_extended (52g)
POU2F1_extended (70g)
TCF12 (12g)
CEBPE_extended (27g)
FOSB_extended (28g)
KLF12 (50g)
ZNF518A_extended (16g)
ZNF160 (41g)
SOX4_extended (71g)
THAP11 (16g)
Regulon activity
4
2
0
−2
−4
CellType
C1 – AM
C2 – AM
C3 – AM
C4 – MoMac
C5 – Neu
C6 – Neu
C7 – LT
C8 – LT
C9 – Eos
C10 – Other
C11 – Mast

```r
sessionInfo()
```

```
## R version 4.3.2 (2023-10-31)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.3 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=fr_BE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=fr_BE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=fr_BE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=fr_BE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Brussels
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] data.table_1.14.8     BiocParallel_1.34.2   RcisTarget_1.20.0
##  [4] tidyHeatmap_1.8.1     stringr_1.5.0         SCENIC_1.3.1
##  [7] ComplexHeatmap_2.16.0 ggplot2_3.4.2         patchwork_1.1.2
## [10] SeuratObject_4.1.3    Seurat_4.3.0          dplyr_1.1.2
##
## loaded via a namespace (and not attached):
##   [1] RcppAnnoy_0.0.21            splines_4.3.2
##   [3] later_1.3.1                 bitops_1.0-7
##   [5] R.oo_1.25.0                 tibble_3.2.1
##   [7] polyclip_1.10-4             graph_1.78.0
##   [9] XML_3.99-0.14               lifecycle_1.0.3
##  [11] doParallel_1.0.17           processx_3.8.1
##  [13] globals_0.16.2              lattice_0.22-5
##  [15] MASS_7.3-60                 crosstalk_1.2.0
##  [17] dendextend_1.17.1           magrittr_2.0.3
##  [19] plotly_4.10.2               rmarkdown_2.23
##  [21] yaml_2.3.7                  httpuv_1.6.11
##  [23] sctransform_0.3.5           spam_2.9-1
##  [25] sp_2.0-0                    spatstat.sparse_3.0-2
##  [27] reticulate_1.30             chromote_0.1.2
##  [29] cowplot_1.1.1               pbapply_1.7-2
##  [31] DBI_1.1.3                   RColorBrewer_1.1-3
##  [33] abind_1.4-5                 zlibbioc_1.46.0
##  [35] GenomicRanges_1.52.0        Rtsne_0.16
##  [37] R.utils_2.12.2              purrr_1.0.1
##  [39] BiocGenerics_0.46.0         RCurl_1.98-1.12
##  [41] webshot2_0.1.1              circlize_0.4.15
```

```
##  [43] GenomeInfoDbData_1.2.10     IRanges_2.34.0
##  [45] S4Vectors_0.38.1            ggrepel_0.9.3
##  [47] irlba_2.3.5.1               listenv_0.9.0
##  [49] spatstat.utils_3.0-3        goftest_1.2-3
##  [51] spatstat.random_3.1-5       annotate_1.78.0
##  [53] fitdistrplus_1.1-11         parallelly_1.36.0
##  [55] DelayedMatrixStats_1.22.1   leiden_0.4.3
##  [57] codetools_0.2-19            DelayedArray_0.26.3
##  [59] tidyselect_1.2.0            shape_1.4.6
##  [61] farver_2.1.1                viridis_0.6.3
##  [63] matrixStats_1.0.0           stats4_4.3.2
##  [65] spatstat.explore_3.2-1      jsonlite_1.8.7
##  [67] GetoptLong_1.0.5            ellipsis_0.3.2
##  [69] progressr_0.13.0            ggridges_0.5.4
##  [71] survival_3.5-7              iterators_1.0.14
##  [73] foreach_1.5.2               tools_4.3.2
##  [75] ica_1.0-3                   Rcpp_1.0.11
##  [77] glue_1.6.2                  gridExtra_2.3
##  [79] xfun_0.39                   MatrixGenerics_1.12.2
##  [81] websocket_1.4.1             GenomeInfoDb_1.36.0
##  [83] AUCell_1.22.0               withr_2.5.0
##  [85] fastmap_1.1.1               fansi_1.0.4
##  [87] digest_0.6.33               R6_2.5.1
##  [89] mime_0.12                   colorspace_2.1-0
##  [91] scattermore_1.2             tensor_1.5
##  [93] spatstat.data_3.0-1         RSQLite_2.3.1
##  [95] R.methodsS3_1.8.2           utf8_1.2.3
##  [97] tidyr_1.3.0                 generics_0.1.3
##  [99] S4Arrays_1.0.4              httr_1.4.6
## [101] htmlwidgets_1.6.2           uwot_0.1.16
## [103] pkgconfig_2.0.3             gtable_0.3.3
## [105] blob_1.2.4                  lmtest_0.9-40
## [107] XVector_0.40.0              htmltools_0.5.5
## [109] dotCall64_1.0-2             GSEABase_1.62.0
## [111] clue_0.3-64                 scales_1.2.1
## [113] Biobase_2.60.0              png_0.1-8
## [115] knitr_1.43                  rstudioapi_0.14
## [117] reshape2_1.4.4              rjson_0.2.21
## [119] nlme_3.1-163                zoo_1.8-12
## [121] cachem_1.0.8                GlobalOptions_0.1.2
## [123] KernSmooth_2.23-22          parallel_4.3.2
## [125] miniUI_0.1.1.1              arrow_12.0.1
## [127] AnnotationDbi_1.62.1        pillar_1.9.0
## [129] vctrs_0.6.3                 RANN_2.6.1
## [131] promises_1.2.0.1            xtable_1.8-4
## [133] cluster_2.1.4               evaluate_0.21
## [135] magick_2.7.5                cli_3.6.1
## [137] compiler_4.3.2              rlang_1.1.1
## [139] crayon_1.5.2                future.apply_1.11.0
## [141] labeling_0.4.2              ps_1.7.5
## [143] plyr_1.8.8                  stringi_1.7.12
## [145] viridisLite_0.4.2           deldir_1.0-9
## [147] assertthat_0.2.1            munsell_0.5.0
## [149] Biostrings_2.68.1           lazyeval_0.2.2
```

```
## [151] spatstat.geom_3.2-4         Matrix_1.6-1
## [153] sparseMatrixStats_1.12.0    bit64_4.0.5
## [155] future_1.33.0               KEGGREST_1.40.0
## [157] shiny_1.7.4.1               highr_0.10
## [159] SummarizedExperiment_1.30.2 ROCR_1.0-11
## [161] igraph_1.5.0.1              memoise_2.0.1
## [163] bit_4.0.5
```